

SMART INDIA HACKATHON 2026

Official idea-submission title page (template structure preserved)

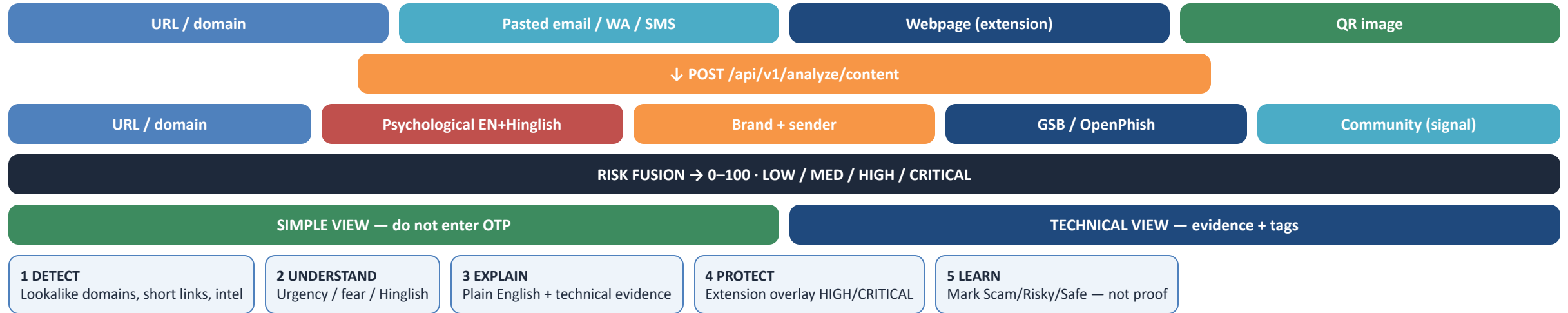
Field	Value
Problem Statement ID	SIH1454
Problem Statement Title	Create an intelligent system using AI/ML to detect phishing domains which imitate look and feel of genuine domains
Theme	Blockchain & Cybersecurity
PS Category	Software
Team ID	[TEAM ID] — fill from SIH portal
Team Name (Registered on portal)	[TEAM NAME] — fill from SIH portal

Organisation (PS): National Technical Research Organisation (NTRO)

IDEA TITLE

PhishShield AI — Analyze before you trust

Proposed Solution · How it addresses the problem · Innovation and uniqueness



Honesty: paste not inbox · OCR partial · native SMS planned · trained CNN planned · confidence = signal agreement, not accuracy

TECHNICAL APPROACH

Technologies to be used · Methodology / process · Working prototype



Actual architecture (not Flask, not a trained CNN)



Prototype: React :5173 + FastAPI :8000 + unpacked Chrome extension. Key files: content_analyzer.py, risk_engine.py, url_checker.py, psychology.py, Scanner.jsx, extension/

NOT in repo as live: Flask · PostgreSQL server · TensorFlow CNN · sklearn model · Gmail OAuth · Android SMS

FEASIBILITY AND VIABILITY

Analysis of feasibility · Challenges and risks · Strategies

TECHNICAL
Working FastAPI+React prototype · pytest · modular services · OpenCV QR in requirements

OPERATIONAL
Paste workflow · extension on current tab · Simple View · no private inbox required

SECURITY
IMPLEMENTED: bcrypt, JWT, SSRF, report caps. PARTIAL: CORS *. PLANNED: global API rate limit

Challenge → mitigation

Challenge	Mitigation
Marketing language false positives	Psychology cannot raise HIGH/CRITICAL alone
New domains / shorteners	RDAP age + unwrap + GSB/OpenPhish + brand distance
Community report abuse	Login + 1 URL/24h + 20/day
Privacy / inbox access	User-pasted content only
SQLite at multi-server scale	PLANNED PostgreSQL
No labelled CNN yet	PLANNED model; explainable fusion today

IMPACT AND BENEFITS

Target audience · Social / economic benefits · Proposed metrics (not claimed results)

Elderly users

First-time UPI / banking

Students

Everyday smartphone users

Non-technical staff

BEFORE

Suspicious link → padlock / jargon → user does not understand → enters OTP on fake bank

WITH PHISHSHIELD

Same link scanned → fusion score → “This may be pretending to be SBI/PayPal. Do not enter OTP.” → go back or continue consciously

Proposed evaluation metrics — NOT claimed F1/accuracy (none published in this repo)

Detection recall on lookalikes

False-positive rate on real banks

Median analysis latency

Usable Simple View on HIGH/CRITICAL

Community report precision after review

Coverage: URL / paste / QR / extension

RESEARCH AND REFERENCES

Details / links of reference and research work actually used

Source	Why it is here
SIH1454 / NTRO	Official problem: phishing domains imitating genuine domains
Google Safe Browsing API v4	Optional live IOC lookup
OpenPhish	Fallback phishing URL feed
RDAP (RFC 7480+)	Domain age for newly registered lookalikes
OWASP SSRF / phishing	Informed private-IP blocking
FastAPI / React / Vite	Actual web stack (not Flask)
Chrome Extensions MV3	Service worker + content scripts
SQLite	Live database
OpenCV	QR decode

No fabricated papers or accuracy numbers. Fill Team ID / Team Name on slide 1 before portal upload. Portal accepts PDF.